QS-VC

Quantum-Safe Video Conferencing

Secure HD/4K video calls with NIST-approved post-quantum cryptography, real-time AI captions and translation, and flexible deployment — ready for the quantum era.

PQC L5	10K+	99+	99.995%
NIST SECURITY LEVEL	PARTICIPANTS/MEETING	LANGUAGES STT	UPTIME SLA

End-to-End Encrypted · Zero-Knowledge SFU · AI-Powered Captions · Air-Gap Ready

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Why Quantum-Safe Video Conferencing?

Today's encrypted video calls use RSA/ECC algorithms that quantum computers will break. Adversaries are already harvesting encrypted traffic now — the "Harvest Now, Decrypt Later" threat. QS-VC protects your calls with NIST-approved post-quantum cryptography so every frame, every word, and every screen share remains confidential — forever.

VC FEATURES AT A GLANCE

HD/4K Video
VP9/AV1 codecs, Simulcast + SVC layers, adaptive bitrate, up to 4K@30fps quality.

Crystal Audio
Opus 48kHz stereo, AI noise suppression (RNNoise + U-Net), echo cancellation.

Screen Sharing
Full-screen & application sharing, real-time annotation, remote control.

Whiteboard
CRDT-based collaborative whiteboard with drawing, text, shapes, infinite canvas.

Recording
Cloud & on-premise, MP4 composite, per-participant tracks, MinIO/S3.

Live Streaming
RTMP ingest to HLS/DASH output, CDN distribution, 10K–100K audience broadcast.

Smart Camera
Auto-framing, person tracking, group mode, whiteboard detection via MediaPipe.

Meeting Tools
Breakout rooms, waiting lobby, raise hand, reactions, polls, mute management.

Interop
SIP/H.323 gateway for legacy room systems, PSTN dial-in fallback.

AI REAL-TIME PIPELINE (<300MS LATENCY)

Capture — Opus 48kHz audio → **Denoise** — RNNoise + U-Net → **Transcribe** — Whisper (99+ langs) → **Translate** — NLLB-200 subtitles → **Display** — Real-time captions

Quantum-Safe Security Architecture

Every layer of QS-VC's media pipeline is protected with post-quantum cryptography — from signaling to media frames to recordings at rest.

1. **Signaling Security (Quantum-TLS)**
   Hybrid Kyber-1024 + X25519 key exchange, Dilithium-5 + Ed25519 signatures, TLS 1.3 with PQC extensions
2. **Media End-to-End Encryption (E2EE)**
   SFrame via Insertable Streams, per-participant Kyber keypairs, frame-level AES-256-GCM, zero-knowledge SFU
3. **Key Management**
   HSM-backed (Shamir 5-of-8), 60-second media key rotation, hourly TURN credential rotation
4. **Recording Encryption**
   Kyber-wrapped Data Encryption Keys, encrypted at-rest storage (MinIO/S3), tamper-proof metadata
5. **Audit Trail**
   Merkle-tree verified hash chains, Dilithium-signed audit entries, configurable retention

DEPLOYMENT MODELS

☁️ SaaS Cloud

- Multi-tenant Kubernetes (EKS/GKE)
- Auto-scaling SFU pools
- Region-selectable data residency
- Zero-downtime rolling updates
- Per-user/month subscription

🏢 On-Premise (Air-Gapped)

- 100% on-site — zero internet needed
- HSM integration (Thales Luna)
- Local AI inference (NVIDIA A10 GPU)
- K3s/RKE2 lightweight Kubernetes
- 4–8 hour automated installation

🔗 Hybrid

- Edge media nodes + cloud orchestration
- Media stays local, control in cloud
- Direct edge-to-edge media routing
- 72-hour offline grace period
- Auto state sync on reconnect

SCALABILITY

📊 Meeting Size Routing

- 1–49 users: Single SFU pod
- 50–500: SFU cascade (3–10 pods)
- 500–10K (Webinar): SFU + MCU hybrid + CDN
- 10K–100K (Broadcast): MCU + HLS/DASH

⚡ Low-Bandwidth Optimization

- >2 Mbps: 1080p/30fps, full AI
- 500K–2M: 360p/30fps, captions
- 100K–500K: 180p/15fps, active speaker
- <100K: Audio-only + PSTN fallback

Plans & Pricing

Choose the right plan for your video conferencing needs.

Business

SaaS | Per-user/month

- ✓ Up to 300 participants per call
- ✓ HD 1080p video & audio
- ✓ Screen sharing & recording
- ✓ AI captions (English)
- ✓ Basic noise suppression
- ✓ SSO (SAML 2.0 / OIDC)
- ✓ Email support

Enterprise

SaaS/Hybrid | Custom Pricing

- ✓ Up to 10,000 participants
- ✓ 4K video + live streaming
- ✓ AI: 99+ language STT & translation
- ✓ Quantum-safe E2EE (PQC)
- ✓ Meeting summarization & actions
- ✓ SIP/H.323 room system gateway
- ✓ Whiteboard & breakout rooms
- ✓ Dedicated CSM + 24/7 support

Sovereign

On-Premise | Perpetual + AMC

- ✓ Air-gapped deployment
- ✓ HSM key management
- ✓ All AI features (local GPU)
- ✓ Full data sovereignty
- ✓ Custom branding / white-label
- ✓ LDAP / Active Directory
- ✓ On-site deployment team
- ✓ Offline operation (72hr grace)

Secure Your Communications Today

Schedule a personalized demo and experience quantum-safe video conferencing that protects your calls against today's and tomorrow's threats.

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